

PURPLE COMPANY – E-COMMERCE ORDER MANAGEMENT

DBMS – ORDER AND ORDER_ITEM MANAGEMENT

1. DATABASE DESIGN

1.1 ORDER TABLE

Attribute	Description
Order_ID	Unique Order ID
Customer_ID	Customer ID
Order_Date	Date of order
Total_Amount	Total amount
Order_Status	Order status

1.2 ORDER_ITEM TABLE

Attribute	Description
Order_Item_ID	Unique Order Item ID
Order_ID	Order ID
Product_ID	Product ID
Quantity	Quantity ordered
Price	Product price

2. RELATIONSHIP

ORDER → ORDER_ITEM

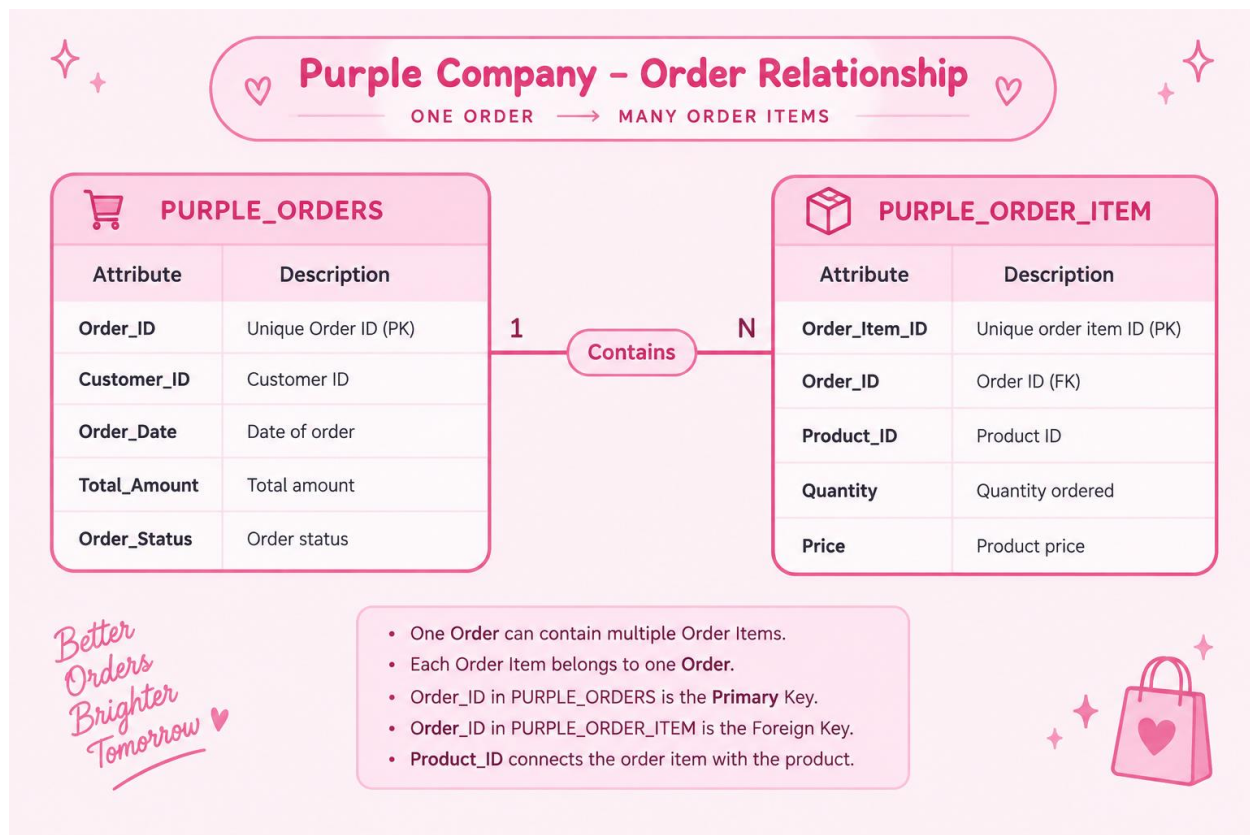
One Order can contain multiple Order Items.

Each Order Item belongs to one Order.

Order_ID in the ORDER table is the Primary Key.

Order_ID in the ORDER_ITEM table is the Foreign Key.

Product_ID connects the order item with the product.



3. TABLE CREATION

3.1 ORDER TABLE

```
SQL> CREATE TABLE PURPLE_ORDERS (
2     Order_ID INT PRIMARY KEY,
3     Customer_ID INT,
4     Order_Date DATE,
5     Total_Amount DECIMAL(10,2),
6     Order_Status VARCHAR2(50)
7 );
```

Table created.

3.2 ORDER_ITEM TABLE

```
SQL> CREATE TABLE PURPLE_ORDER_ITEM (  
2      Order_Item_ID INT PRIMARY KEY,  
3      Order_ID INT,  
4      Product_ID INT,  
5      Quantity INT,  
6      Price DECIMAL(10,2),  
7      FOREIGN KEY (Order_ID)  
8      REFERENCES PURPLE_ORDERS(Order_ID)  
9  );
```

4. ORDER TABLE DATA INSERTION

```
SQL>  
SQL> INSERT INTO PURPLE_ORDERS  
2  (Order_ID, Customer_ID, Order_Date, Total_Amount, Order_Status)  
3  VALUES  
4  (103, 3, '22-SEP-2026', 3200.00, 'Shipped');  
  
1 row created.  
  
SQL>  
SQL> INSERT INTO PURPLE_ORDERS  
2  (Order_ID, Customer_ID, Order_Date, Total_Amount, Order_Status)  
3  VALUES  
4  (104, 4, '23-SEP-2026', 1500.00, 'Pending');  
  
1 row created.  
  
SQL>  
SQL> INSERT INTO PURPLE_ORDERS  
2  (Order_ID, Customer_ID, Order_Date, Total_Amount, Order_Status)  
3  VALUES  
4  (105, 5, '23-SEP-2026', 4500.00, 'Delivered');  
  
1 row created.  
  
SQL>  
SQL> INSERT INTO PURPLE_ORDERS  
2  (Order_ID, Customer_ID, Order_Date, Total_Amount, Order_Status)  
3  VALUES  
4  (106, 6, '24-SEP-2026', 2750.00, 'Processing');  
  
1 row created.  
  
SQL>  
SQL> COMMIT;  
  
Commit complete.  
  
SQL> SELECT * FROM PURPLE_ORDERS;
```

4.1 DISPLAY ORDER TABLE DATA

```
SQL> SELECT * FROM PURPLE_ORDERS;
```

ORDER_ID	CUSTOMER_ID	ORDER_DAT	TOTAL_AMOUNT	ORDER_STATUS
101	1	20-SEP-26	2500	Delivered
102	2	21-SEP-26	1800	Processing
103	3	22-SEP-26	3200	Shipped
104	4	23-SEP-26	1500	Pending
105	5	23-SEP-26	4500	Delivered
106	6	24-SEP-26	2750	Processing

5. ORDER_ITEM TABLE DATA INSERTION

```
SQL> INSERT INTO PURPLE_ORDER_ITEM
2 (Order_Item_ID, Order_ID, Product_ID, Quantity, Price)
3 VALUES
4 (2, 102, 202, 1, 1800.00);
INSERT INTO PURPLE_ORDER_ITEM
*
ERROR at line 1:
ORA-00001: unique constraint (SYSTEM.SYS_C008461) violated

SQL>
SQL> INSERT INTO PURPLE_ORDER_ITEM
2 (Order_Item_ID, Order_ID, Product_ID, Quantity, Price)
3 VALUES
4 (3, 103, 203, 2, 1600.00);
1 row created.

SQL>
SQL> INSERT INTO PURPLE_ORDER_ITEM
2 (Order_Item_ID, Order_ID, Product_ID, Quantity, Price)
3 VALUES
4 (4, 104, 204, 1, 1500.00);
1 row created.

SQL>
SQL> INSERT INTO PURPLE_ORDER_ITEM
2 (Order_Item_ID, Order_ID, Product_ID, Quantity, Price)
3 VALUES
4 (5, 105, 205, 3, 1500.00);
1 row created.

SQL>
SQL> INSERT INTO PURPLE_ORDER_ITEM
2 (Order_Item_ID, Order_ID, Product_ID, Quantity, Price)
3 VALUES
4 (6, 106, 206, 2, 1375.00);
1 row created.
```

5.1 DISPLAY ORDER_ITEM TABLE DATA

6 rows selected.

```
SQL> SELECT * FROM PURPLE_ORDER_ITEM;
```

ORDER_ITEM_ID	ORDER_ID	PRODUCT_ID	QUANTITY	PRICE
1	101	201	2	1000
2	102	202	1	1800
3	103	203	2	1600
4	104	204	1	1500
5	105	205	3	1500
6	106	206	2	1375

6 rows selected.